

**DYNAMICAL SYSTEMS:
CLASS SEPTEMBER 10, 2026
5B. 2D LINEAR SYSTEMS: CLASSIFICATION OF FIXED POINTS**

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Teams

Post screenshots of your work to the course Google Drive today: Link to drive folder. Include words, labels, and other short notes on your whiteboard that might make those solutions useful to you or your classmates. Name your images using the format:

- teamXX-YY.EXT

where XX is your two-digit team number (e.g. 06 if you are team 6), YY is the two digit number of image for today (e.g. 01 for the first image, 02 for the second, and so on.), and EXT is the extension for the image filetype you used.

Team Activity

1. Starting with your roles at the end of last time, rotate.

(6.3.6) Consider the system

$$\begin{aligned}\dot{x} &= f(x, y) = xy - 1 \\ \dot{y} &= x - y^3\end{aligned}$$

- (a) Use **substitution** to show that $(-1, -1)$ and $(1, 1)$ are both fixed points of the system (i.e. is $f(x, y) = 0$ and $g(x, y) = 0$ at these points?)
- (b) Use Taylor polynomials to approximate the dynamical system to the second order about the fixed point $(-1, -1)$. Let $u = x - (-1)$, $v = y - (-1)$ and use this to simplify your expressions.

I am asking you to approximate to second order as a review of Taylor approximation.

Extra note on Taylor polynomials:

A linear approximation to a function at a point Q has the same value as the function of interest at Q and has the same first derivatives as the original function at Q . A higher order approximation, of order p has the same value and derivatives, up to the order p derivative, as the original function at Q .

It may be helpful to recall that $f(x, y) \approx f(a, b) + (x - a)f_x(a, b) + (y - b)f_y(a, b) + \frac{1}{2}(x - a)^2 f_{xx}(a, b) + (x - a)(y - b)f_{xy}(a, b) + \frac{1}{2}(y - b)^2 f_{yy}(a, b) + h.o.t$

- (c) Sufficiently close to $(-1, -1)$, we have $|u|, |v| \ll 1$ and $u^2 \ll |u|$, $v^2 \ll |v|$, so quadratic order and higher terms are small relative to the linear terms.
 - Drop these higher order terms to generate a linearization of the system/
 - Use your linearization to write a dynamical system of the form

$$\underline{\dot{u}} = A\underline{u}$$

giving definitions for $\underline{u}$, A .

- Explain why the linearization leads to this kind of matrix equation only at a fixed point. What would the form of the linearized system away from a fixed point?
- (d) Create a linearized system about the fixed point $(1, 1)$ as well.
- (e) Classify your points as **hyperbolic** (*no eigenvalues have zero real part*) or **nonhyperbolic** (*at least one eigenvalue has zero real part*).

Since $\Delta = \lambda_1 \lambda_2$, there must be a zero eigenvalue when $\Delta = 0$. If $\tau = 0$ there may be a complex conjugate pair of eigenvalues with zero real part, the eigenvalues might both be real and sum to zero, or the eigenvalues might both be zero. In the case where a c.c. pair with zero real part, find the sign of the determinant.

The Hartman-Grobman theorem tells us that stability information from the linearization can be used to classify hyperbolic fixed points. When a fixed point is non-hyperbolic the stability information from linearization is not so useful.

Identify the stability of any hyperbolic fixed points. (Classify them as attracting, repelling, or saddle points, and identify whether they are stable or unstable).

- (f) use eigenvalues and eigenvectors to sketch neighbouring trajectories to any fixed points with real eigenvalues. Try to fill in the rest of the phase portrait.

Solution. We are given the nonlinear system

$$\dot{x} = f(x, y) = xy - 1, \quad \dot{y} = g(x, y) = x - y^3.$$

- (a) We first verify that $(-1, -1)$ and $(1, 1)$ are fixed points.

At $(-1, -1)$,

$$f(-1, -1) = (-1)(-1) - 1 = 1 - 1 = 0$$

and

$$g(-1, -1) = -1 - (-1)^3 = -1 + 1 = 0.$$

Therefore $(-1, -1)$ is a fixed point.

Similarly, at $(1, 1)$,

$$f(1, 1) = (1)(1) - 1 = 0$$

and

$$g(1, 1) = 1 - 1^3 = 0.$$

Therefore $(1, 1)$ is also a fixed point.

- (b) We now construct a second-order Taylor approximation about the fixed point $(-1, -1)$. Define

$$u = x - (-1) = x + 1, \quad v = y - (-1) = y + 1,$$

so that

$$x = -1 + u, \quad y = -1 + v.$$

Since u and v differ from x and y only by constants,

$$\dot{u} = \dot{x}, \quad \dot{v} = \dot{y}.$$

Substituting into the first equation gives

$$\begin{aligned} \dot{u} &= (-1 + u)(-1 + v) - 1 \\ &= 1 - u - v + uv - 1 \\ &= -u - v + uv. \end{aligned}$$

Thus the second-order Taylor polynomial for the first component is actually exact:

$$\dot{u} = -u - v + uv.$$

For the second equation,

$$\dot{v} = (-1 + u) - (-1 + v)^3.$$

Since

$$(-1 + v)^3 = -1 + 3v - 3v^2 + v^3,$$

we obtain

$$\begin{aligned}\dot{v} &= (-1 + u) - (-1 + 3v - 3v^2 + v^3) \\ &= u - 3v + 3v^2 - v^3.\end{aligned}$$

Keeping terms through second order gives

$$\dot{v} \approx u - 3v + 3v^2.$$

Hence the second-order approximation about $(-1, -1)$ is

$$\begin{aligned}\dot{u} &= -u - v + uv, \\ \dot{v} &= u - 3v + 3v^2.\end{aligned}$$

This can also be obtained directly from the Taylor derivatives. The Jacobian of the original system is

$$J(x, y) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix} = \begin{pmatrix} y & x \\ 1 & -3y^2 \end{pmatrix}.$$

At $(-1, -1)$,

$$J(-1, -1) = \begin{pmatrix} -1 & -1 \\ 1 & -3 \end{pmatrix}.$$

(c) Sufficiently close to the fixed point, $|u|, |v| \ll 1$, and therefore the quadratic and higher-order terms are small compared with the linear terms. Dropping uv , $3v^2$, and all higher-order terms gives

$$\begin{aligned}\dot{u} &= -u - v, \\ \dot{v} &= u - 3v.\end{aligned}$$

Thus

$$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}.$$

In the notation

$$\underline{u} = \begin{pmatrix} u \\ v \end{pmatrix}, \quad A = \begin{pmatrix} -1 & -1 \\ 1 & -3 \end{pmatrix},$$

this is

$$\dot{\underline{u}} = A\underline{u}.$$

The reason that the linearization has the homogeneous form

$$\dot{\underline{u}} = A\underline{u}$$

is that we are expanding about a fixed point. For a general nonlinear system

$$\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}),$$

and a point $\mathbf{x}_0$, let

$$\mathbf{u} = \mathbf{x} - \mathbf{x}_0.$$

The first-order Taylor approximation is

$$\dot{\mathbf{u}} \approx \mathbf{F}(\mathbf{x}_0) + D\mathbf{F}(\mathbf{x}_0)\mathbf{u}.$$

At a fixed point,

$$\mathbf{F}(\mathbf{x}_0) = \mathbf{0},$$

so the constant term disappears, leaving

$$\dot{\mathbf{u}} = D\mathbf{F}(\mathbf{x}_0)\mathbf{u}.$$

Away from a fixed point, the linearized system is instead an *affine* system of the form

$$\dot{\mathbf{u}} = \mathbf{F}(\mathbf{x}_0) + D\mathbf{F}(\mathbf{x}_0)\mathbf{u}.$$

(d) We now linearize about the other fixed point $(1, 1)$.

Let

$$u = x - 1, \quad v = y - 1,$$

so that

$$x = 1 + u, \quad y = 1 + v.$$

Then

$$\begin{aligned} \dot{u} &= (1 + u)(1 + v) - 1 \\ &= u + v + uv, \end{aligned}$$

while

$$\begin{aligned} \dot{v} &= (1 + u) - (1 + v)^3 \\ &= (1 + u) - (1 + 3v + 3v^2 + v^3) \\ &= u - 3v - 3v^2 - v^3. \end{aligned}$$

Dropping quadratic and higher-order terms gives

$$\begin{aligned} \dot{u} &= u + v, \\ \dot{v} &= u - 3v. \end{aligned}$$

Therefore the linearized system at $(1, 1)$ is

$$\begin{pmatrix} \dot{u} \\ \dot{v} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & -3 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}.$$

Notice that this agrees with the Jacobian evaluated at $(1, 1)$:

$$J(1, 1) = \begin{pmatrix} 1 & 1 \\ 1 & -3 \end{pmatrix}.$$

(e) We classify the two fixed points using the eigenvalues of their linearizations.

At $(-1, -1)$:

We have

$$A_- = \begin{pmatrix} -1 & -1 \\ 1 & -3 \end{pmatrix}.$$

The trace and determinant are

$$\tau = -1 - 3 = -4, \quad \Delta = (-1)(-3) - (-1)(1) = 4.$$

The characteristic polynomial is

$$\begin{aligned} \det(A_- - \lambda I) &= \begin{vmatrix} -1 - \lambda & -1 \\ 1 & -3 - \lambda \end{vmatrix} \\ &= (-1 - \lambda)(-3 - \lambda) + 1 \\ &= \lambda^2 + 4\lambda + 4 \\ &= (\lambda + 2)^2. \end{aligned}$$

Therefore

$$\lambda_1 = \lambda_2 = -2.$$

Both eigenvalues have strictly negative real part, so $(-1, -1)$ is

hyperbolic.

Because both eigenvalues are negative, trajectories sufficiently close to the fixed point move toward it as $t \rightarrow \infty$. Therefore $(-1, -1)$ is an

attracting, asymptotically stable fixed point.

Because the repeated eigenvalue has only one linearly independent eigenvector, the fixed point is more specifically a

stable improper node.

To see this, solve

$$(A_- + 2I)\mathbf{q} = 0.$$

Since

$$A_- + 2I = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix},$$

we obtain

$$q_1 = q_2.$$

Thus an eigenvector is

$$\mathbf{q} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

At $(1, 1)$:

Here

$$A_+ = \begin{pmatrix} 1 & 1 \\ 1 & -3 \end{pmatrix}.$$

Its trace and determinant are

$$\tau = 1 - 3 = -2, \quad \Delta = (1)(-3) - (1)(1) = -4.$$

Its characteristic polynomial is

$$\begin{aligned}\det(A_+ - \lambda I) &= \begin{vmatrix} 1 - \lambda & 1 \\ 1 & -3 - \lambda \end{vmatrix} \\ &= (1 - \lambda)(-3 - \lambda) - 1 \\ &= \lambda^2 + 2\lambda - 4.\end{aligned}$$

Hence

$$\lambda = \frac{-2 \pm \sqrt{4 + 16}}{2} = -1 \pm \sqrt{5}.$$

Thus

$$\lambda_1 = -1 + \sqrt{5} > 0, \quad \lambda_2 = -1 - \sqrt{5} < 0.$$

Neither eigenvalue has zero real part, so $(1, 1)$ is also

hyperbolic.

Since one eigenvalue is positive and the other is negative, $(1, 1)$ is a

saddle point and therefore unstable.

For completeness, if a two-dimensional system has a complex conjugate pair with zero real part, then

$$\lambda_{1,2} = \pm i\omega.$$

Their product is

$$\lambda_1 \lambda_2 = (i\omega)(-i\omega) = \omega^2 > 0.$$

Therefore, when the eigenvalues are a nonzero pure imaginary conjugate pair,

$$\tau = 0, \quad \Delta > 0.$$

(f) We use the eigenvectors to determine the local phase portrait.

Near $(-1, -1)$:

The repeated eigenvalue is

$$\lambda = -2$$

with eigenvector

$$\mathbf{q} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Thus the eigendirection in (u, v) coordinates is

$$v = u.$$

Since

$$u = x + 1, \quad v = y + 1,$$

this corresponds to

$$y + 1 = x + 1,$$

or simply

$$y = x.$$

Because the eigenvalue is negative, arrows along this eigendirection point *toward* $(-1, -1)$. Since the matrix has only one independent eigenvector, nearby trajectories curve into the

fixed point and become tangent to the line $y = x$ as they approach it. Thus $(-1, -1)$ appears as an attracting improper node.

Near $(1, 1)$:

For the positive eigenvalue

$$\lambda_u = -1 + \sqrt{5},$$

solve

$$(1 - \lambda_u)u + v = 0.$$

Hence

$$v = (\lambda_u - 1)u = (\sqrt{5} - 2)u.$$

An unstable eigenvector is therefore

$$\mathbf{q}_u = \begin{pmatrix} 1 \\ \sqrt{5} - 2 \end{pmatrix}.$$

Since $\lambda_u > 0$, trajectories move *away* from the saddle along this direction.

For the negative eigenvalue

$$\lambda_s = -1 - \sqrt{5},$$

we obtain

$$v = (\lambda_s - 1)u = -(2 + \sqrt{5})u.$$

Thus a stable eigenvector is

$$\mathbf{q}_s = \begin{pmatrix} 1 \\ -2 - \sqrt{5} \end{pmatrix}.$$

Since $\lambda_s < 0$, trajectories move *toward* $(1, 1)$ along this eigendirection.

Therefore the approximate stable and unstable lines through $(1, 1)$ are

$$y - 1 = -(2 + \sqrt{5})(x - 1) \quad (\text{stable})$$

and

$$y - 1 = (\sqrt{5} - 2)(x - 1) \quad (\text{unstable}).$$

These lines give the local saddle geometry.

To fill in the remainder of the nonlinear phase portrait, consider the nullclines. The x -nullcline is

$$\dot{x} = 0 \iff xy = 1 \iff \boxed{x = \frac{1}{y}},$$

while the y -nullcline is

$$\dot{y} = 0 \iff x = y^3,$$

so

$$\boxed{x = y^3}.$$

Their intersections satisfy

$$y^3 y = 1,$$

and hence

$$y^4 = 1.$$

The real intersections are exactly

$$(-1, -1) \quad \text{and} \quad (1, 1).$$

The direction of the vector field in the regions separated by the nullclines is determined from

$$\dot{x} = xy - 1 \begin{cases} > 0, & xy > 1, \\ < 0, & xy < 1, \end{cases}$$

and

$$\dot{y} = x - y^3 \begin{cases} > 0, & x > y^3, \\ < 0, & x < y^3. \end{cases}$$

These sign rules, together with the eigendirections above, can be used to sketch the complete phase portrait.

In summary,

Fixed point	Eigenvalues	Type	Stability
$(-1, -1)$	$-2, -2$	improper node	attracting/stable
$(1, 1)$	$-1 + \sqrt{5}, -1 - \sqrt{5}$	saddle	unstable

and both fixed points are hyperbolic.

□